Name: ____________________ **Student ID:** ______________

1. (10 points) Kelsey's Marshallian demand for x is given by

$$g_x(p_x, p_y, I) = \frac{I - 2p_x}{3p_y}$$

Her income is $I = 20$, and she faces prices $p_x = 4$ and $p_y = 1$. What is the income elasticity of demand for y?

2. (10 points) Consider the following production function:

$$q = L^{1/3}K^{2/3}$$

Using factor prices v for capital and w for labor, solve for **one** of the contingent demands, either L^c or K^c.

Bonus (+1 point): Find both contingent demands.

3. (15 points) Consider the following utility function:

$$U(X, Y) = 2XY$$

   (a) Solve for the Marshallian demands g_x and g_y.

   (b) Income is $I = 4$ and prices are $p_x = \frac{2}{3}$ and $p_y = 1$. What are the optimal demands and level of indirect utility?

   (c) Suppose there is a tax $T = 2$ on X (new price $= p_x + T$). How much additional income would a consumer need to receive to be as well-off as before the tax?

Bonus: What is the correct spelling of your Econ 11 Professor's last name?